Ask Small VLMs Only What They Can See: A Semantic-to-Metric Agent for Embodied Navigation

Supplementary Material

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Paper ID *****

Table S1. Parameters of S2M on EB-Navigation.

Group	Parameter	Value
Questions	presence threshold for named targets	0.5
	crop side (times the box's longer side)	2.5
	crop and tile resolution	448 px
	crop acceptance	0.5
	weight of a rejected crop	$0.2\,\pi_t^z$
Lifting	depth window (fraction of the box)	0.5
	maximum range	6 m
	floor / obstacle height	0.10 / 0.1–1.9 m
	occupancy grid	5 cm
Memory	association radius r	0.5 m
	repeated-view discount γ	0.25
	same view: position, heading	0.2 m, equal
	negative evidence λ	0.4
	image margin, occlusion tolerance	20 px, 0.3 m
	removal threshold	$w_k \le -1$
Search	commit τ_c / keep τ_k	0.8 / 0.4
	approach a weak hypothesis	$w_k \ge 0.35$
	exhaustion radius	0.9 m
	exploration goal distance	0.75–3 m
	tiles, overlap, boxes checked	3×3, 20%, 4
Planning	cost of an unobserved edge point	0.15
	obstacle inflation (fallback)	0.20 (0.10) m
	goal tolerance	0.55 m

S1. Implementation details

Parameters. Table S1 lists every parameter of S2M. All were fixed on the development split (Sec. S2) and are the same for both models and for the RGB-only configuration; RoboTHOR changes only the items of the explicit-stopping paragraph of Sec. 4.4 of the main paper.

Prompts. The target phrase s comes from one text-only query per episode:

```
A robot must navigate to one
object in a house. Instruction:
"⟨ℓ⟩" Reply with JSON {"name":
"...", "appearance_only": true
or false, "looks": "..."} where
- name: the most likely specific
  household object, as its common
  name in 1-2 words (e.g. "toaster",
  "trash can", "laptop"; never a vague
  word like "device" or "object").
- appearance_only: true if the
  instruction describes the object
  only by colour, shape, material
  or texture, without saying what
  it is or what it is used for. -
  looks: the appearance words from
  the instruction (colour, shape,
  material), empty if none.
```

s is the name, or the appearance words when `appearance_only` is true. The two per-frame queries are

```
Does this image show ⟨s⟩? Answer yes
or no.
```

```
Locate the ⟨s⟩ in the image and
output its bounding box as JSON:
[{"bbox_2d": [x1, y1, x2, y2],
"label": "⟨s⟩"}]. If it is not in
the image, output [].
```

The presence query generates one token and reads $P(yes)$ and $P(no)$ from the ten most likely first tokens (Eq. (1)). The crop check asks the presence query and each tile the box query, both on the magnified image. Box replies are parsed with one regular expression; a box narrower than 2 px counts as no box.

$$\pi_t = \frac{\sum_{y \in yes} P(y)}{\sum_{y \in yes} P(y) + \sum_{n \in no} P(n)}, \qquad (1)$$

where the sums run over the case variants of each answer among the ten tokens; $\pi_t = 0.5$ if neither appears.

Serving. Both models run in vLLM with their 4-bit AWQ checkpoints, temperature 0, and at most 96 generated tokens for a box and one for a presence answer. Qwen2.5-VL receives frames resized to at most 409,600 pixels; Qwen3-VL uses its default processor. The two per-frame queries are sent in parallel.

S2. Protocol

Development split. EmbodiedBench provides target objects and the three instruction variants for 90 iTHOR rooms, and its test set uses 60 of them (FloorPlan11–30, 211–230, 311–330). We rerun the benchmark's own generator on the other 30 (FloorPlan1–10, 201–210, 301–310): a start pose 2.5–3.0 m from the first instance of the target with the camera level, Common, Complex, and Visual from the same targets with the benchmark's instruction variants, and Long as Base with the start heading rotated by 180°. This yields five subsets of 30 episodes. Every parameter of Tab. S1 and every design choice of Sec. 4 was selected on this split, and the configuration was frozen before the first test run.

Statistics. All agents run on the same 300 test episodes, so each comparison uses per-episode differences. A configuration run several times enters with its per-episode mean success. Confidence intervals are percentile intervals of 10^4 episode-level bootstrap resamples. p-values are two-sided sign-flip permutation tests with 10^6 random sign assignments; the ten variants of Tab. 2 of the main paper are Holm-corrected together. Comparisons of two single binary runs use the exact McNemar test.

Outcome categories. A failed episode takes the first matching category: the target never enters the view; the final distance is at most 1.5 m (near miss); the target is out of view in the last five steps (lost after seen); at least 30% of the actions are rejected (blocked); at least three immediate reversals (oscillation); otherwise approach failure. Visibility comes from the simulator's object metadata.

RoboTHOR. 17 of the 1800 validation episodes end because the simulator rejects the start pose (`teleport failed`); they are counted as failures.

S3. Additional results

Single views. Table S2 extends the probe of Sec. 3 of the main paper to both models and to the views that do not show the target. Qwen3-VL-4B finds the target more often than Qwen2.5-VL-3B but also places more boxes where there is nothing to find, most of all for appearance-only phrases. Both kinds of error enter the memory, and both are what multi-view fusion and negative evidence act on.

Outcomes. Table S3 gives the outcome of every agent on the 300 test episodes. S2M removes almost all never-seen and blocked episodes with both models. With Qwen2.5-VL-3B the EB prompt loses a target it has seen in 104 episodes; S2M does so in 48, its largest remaining failure mode with the smaller model.

Cost. Per episode, S2M makes 52.6 VLM calls against 7.8 for the EB prompt, but each call asks one short question: it reads 14.4k prompt tokens and generates 513 tokens, against 27.1k and 2.3k for the EB prompt, whose plans and reasoning dominate its generation.

RoboTHOR by category. Table S4 lists success and SPL for each of the twelve categories.

Table S2. Single-view probe on the test episodes of the first four subsets (start view and the three other 90° headings; 75 views with the target visible per subset).

	Box on the target, IoU≥0.3 (%)				Box on a view without it (%)			
Model	Base	Common	Complex	Visual	Base	Common	Complex	Visual
Qwen3-VL-4B	78.7	69.3	77.3	49.3	14.5	18.8	16.4	41.2
Qwen2.5-VL-3B	40.0	33.3	37.3	18.7	3.6	8.5	3.0	8.5

Table S3. Outcomes on EB-Navigation test (episodes of 300; mean over runs where a configuration was run repeatedly).

	Qwen3-VL-4B			Qwen2.5-VL-3B	
Outcome	EB prompt	S2M	S2M, RGB	EB prompt	S2M
success	165	238.2	229.5	61	210
never seen	12	0.8	0	14	0
lost after seen	43	28.5	24.5	104	48
near miss	34	9.8	20.5	26	20
blocked	26	0.5	1	51	3
oscillation	1	10.2	9	1	2
approach failure	19	12	15.5	43	17

Table S4. S2M on RoboTHOR validation by target category.

Category	Episodes	SR	SPL
basketball	150	88.0	54.3
mug	150	77.3	45.7
apple	150	74.7	50.1
garbage can	150	71.3	47.2
television	150	62.7	38.5
baseball bat	150	62.7	34.4
laptop	150	61.3	27.8
house plant	150	56.0	28.7
spray bottle	150	49.3	31.8
alarm clock	150	44.7	29.5
bowl	150	41.3	23.8
vase	150	25.3	18.5
all	1800	59.6	35.9